

Samuel D. Pekofsky

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EDUCATION

University of Wisconsin-Madison

Sep 2021 – May 2025

B.S. in Data Science • Minor in Computer Science • Dean's List

Madison, WI

EXPERIENCE

AI Engineering & Data Consultant

Mar 2026 – Jul 2026

Kubrick Group

London, UK & New York, NY

- Utilized **agile methodologies** to deliver **predictive models** embedded in web apps with **agentic chatbots**.
- Engineered a **RAG pipeline** with **LangChain** that parsed and chunked PDF files into **over 1,000 text segments**, which were then embedded with **OpenAI** and indexed in **ChromaDB**.
- Implemented a **callable retrieval tool** built upon the vector store, alongside a **SQL querying tool**, improving **response relevance and accuracy** on company-specific questions.
- Maintained and **deployed container images to Kubernetes clusters** with Azure.

ML Training Expert (Independent Contractor)

Nov 2024 – Nov 2025

Outlier AI

Remote

- Performed **coding, data annotation, and quality assurance** for large-scale AI projects.
- Analyzed and refined hundreds** of frontier model outputs.

Package Development Intern – clubSandwich

Feb 2025 – Aug 2025

University of Wisconsin-Madison Open Source Program Office

Madison, WI

- Improved an **object-oriented R package (678K+ downloads)** which provides **cluster-robust variance estimators**.
- Addressed GitHub issues, including **implementing highly requested features** and **extending compatibility** to objects from additional packages.
- Consolidated thousands of lines of outdated tests**, contributed new tests, and **updated documentation**.

PROJECTS

Customer Churn Prediction | *Python, scikit-learn, Azure, Jira, Git*

Jun 2026 – Jul 2026

- Utilized **6M+** rows of customer and transaction data to develop **predictive models for customer churn**, with multiple models achieving **F1 scores above 0.7** during testing.
- Designed an **ETL pipeline** to cleanly handle future data with **fitted imputation**.
- Performed **EDA with visualizations and documentation** to justify feature scaling and engineering choices.

Deep Learning for Classifying Images | *Python, PyTorch, torchvision, Git, GitHub*

Mar 2025 – May 2025

- Built a custom **hybrid CNN-Transformer model** and **fine-tuned ResNet-50**, achieving over **98% and 96% accuracy** respectively on image classification despite a **small dataset of 4,782 images**.
- Developed a data preprocessing pipeline that **loads, cleans, and augments images** to balance the dataset.
- Leveraged **grid search** with early stopping to **find optimal hyperparameters**.

US Migration Flows | *R, R Markdown, tidyverse, Shiny, tigris, sf, Leaflet, Git, GitHub*

Feb 2025 – May 2025

- Built an ETL pipeline that ingests migration data and shapefiles via the **U.S. Census Bureau's API**.
- Conducted **multi-year analyses** with extensive data processing, mapping, and **interactive visualizations**.

Twitter Sentiment & Topic Modeling | *Python, pandas, VADER, pyLDAvis, LaTeX*

Oct 2024 – Dec 2024

- Analyzed **2M+ tweets** using **sentiment analysis** and **LDA**, identifying heavily discussed topics from the 2020 election cycle and high variance in sentiment.
- Co-authored a research paper with 10 figures, **documenting sentiment trends** and **topics** surrounding presidential candidates both **domestically and internationally**.

Many additional projects are available at pekofsky.github.io/projects

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, JavaScript, HTML, CSS, LaTeX, Julia, MQL

Libraries & Frameworks: PyTorch, LangChain, scikit-learn, pandas, NumPy, FastAPI, Streamlit, Shiny, Django, Flask, pytest, Selenium, Jekyll, testthat, JUnit, ggplot2, dplyr, devtools

Developer Tools: Git, GitHub, Docker, Kubernetes, Jira, VS Code, uv, RStudio, Google Colab, Eclipse

Platforms & Services: Azure (AKS, DevOps), OpenAI API, ChromaDB, Firebase, Google Cloud Platform